

Altitude Control for a Drone

Control Systems — Information Engineering (Year 3)

October 28, 2025

Problem

You are designing the **vertical (altitude) control** of a small delivery drone that must move from its current height to a new setpoint **smoothly and accurately**, without bouncing the payload and while withstanding a brief wind gust. All modeling and control are done in **continuous time** and implemented in **Simulink**.

At $t = 0$ the altitude reference steps from 0 m to 1 m. At $t = 4$ s the drone is hit by a small upward gust. The controller must meet the performance, actuator, and robustness requirements below.

Given Model (Input–Output Differential Equation)

Let $h(t)$ be altitude (m), $u(t)$ the commanded thrust *after saturation* (N), and $d(t)$ an external vertical force disturbance (N). Gravity is modeled explicitly.

$$m \ddot{h}(t) + b \dot{h}(t) = u(t) - m g + d(t). \quad (1)$$

Measured output: $y(t) = h(t) + n(t)$, where $n(t)$ is optional measurement noise. Reference: $r(t)$ (unit step $0 \rightarrow 1$ m at $t = 0$).

Nominal parameters. $m = 1.0$ kg (robustness tests will vary this), $b = 2.0$ N s/m, $g = 9.81$ m/s².

Disturbance. $d(t) = 0$ for $t < 4$ s; at $t = 4$ s apply a step $d(t) = +0.1$ N (test also -0.1 N).

Note. You may include gravity feedforward (adding $+mg$ inside the controller command before saturation) so the controller mainly regulates deviations about hover. The plant always follows (1).

Specifications & Requirements (Pass/Fail)

Simulation setup. Time horizon: $t \in [0, 10]$ s (continuous-time simulation). The controller structure is your choice. If actuator limits are used, apply saturation directly to the command that drives the plant as $u(t)$.

Performance Summary

- **Static error:** $|e_{ss}^{step}| \leq 0.01$ (i.e., $\leq 1\%$ of the 1 m step).
- **Transient performance (step at $t = 0$):**
 - Percent overshoot $\leq 5\%$.
 - Settling time (within $\pm 2\%$ band) $t_s \leq 3.0$ s.
 - Rise time t_r (10–90%) with $0.5 \text{ s} \leq t_r \leq 1.0 \text{ s}$ (soft target; prioritize overshoot/settling).
- **Actuator constraint:** $|u(t)| \leq 10 \text{ N}$ for all $t \in [0, 10]$ s. If your unsaturated command would exceed this, include *saturation and anti-windup* and still meet the specs above.
- **Disturbance rejection:** For $d(t) = +0.1 \text{ N}$ at $t = 4 \text{ s}$, the output must re-enter and remain within the $\pm 5\%$ band within 1.0 s (i.e., by $t = 5.0 \text{ s}$). Repeat with $d(t) = -0.1 \text{ N}$ and report both results.
- **Robustness:** With mass variation $m \in \{0.8 \text{ kg}, 1.2 \text{ kg}\}$ (i.e., $\pm 20\%$), the closed loop must remain stable and still satisfy $\text{overshoot} \leq 5\%$ and $t_s \leq 3.0 \text{ s}$ for the step at $t = 0$. (Static error should remain $\leq 1\%$ if integral action is used.)
- **Optional noise test:** With $n(t) \sim \mathcal{N}(0, 0.01^2)$ m, verify that static accuracy and transient specs remain satisfied on average over 10 runs with different random seeds.

Deliverables

1. A figure with $y(t)$ and $u(t)$ highlighting the step, overshoot, settling time, and gust recovery.
2. A table reporting: steady-state error, overshoot, rise time, settling time, $\max |u|$, and recovery time after the gust, for nominal and robustness cases; indicate PASS/FAIL.
3. Your controller equations (ODEs), parameter values, and a short paragraph explaining your tuning choices.

Suggested Result Table (Template)

Scenario	Static Error	Overshoot	t_r	t_s	$\max u $	Gust recovery
Nominal ($m = 1.0 \text{ kg}$)						
$m = 0.8 \text{ kg}$						
$m = 1.2 \text{ kg}$						
Gust $d = +0.1 \text{ N}$						
Gust $d = -0.1 \text{ N}$						

Notes for students: For metrics, define overshoot as $(\max y - y_\infty)/y_\infty \times 100\%$, settling time as the first time after which $|y(t) - y_\infty| \leq 0.02 y_\infty$ for all later t , and rise time as the time between 10% and 90% of y_∞ .